The State of Quantum Teleportation Research in 2025: Breakthroughs, Mechanisms, Applications, and Global Leadership

---

Introduction

Quantum teleportation, once the purview of science fiction, has emerged as one of the most transformative and actively advancing frontiers in quantum science and technology. It stands at the heart of the ongoing quantum revolution-a revolution promising to disrupt the foundations of computing, secure communications, and network architectures across the globe. Unlike its fictional namesake, quantum teleportation does not transmit matter instantaneously from one place to another; instead, it enables the transfer of quantum states-or quantum information-between distant parties using entanglement, superposition, and classical communication. This subtle yet profound process unlocks the possibility of virtually unhackable communication, distributed quantum computing, precise quantum sensing, and even new methods for remote energy transfer.

As of mid-2025, quantum teleportation is no longer a laboratory curiosity. Multiple high-profile breakthroughs have been realized, from the Oxford distributed quantum supercomputer using teleported quantum gates, to the demonstration of seamless teleportation across telecom-band networks, and the early commercialization efforts enveloping quantum-encrypted global networks. This report delivers a comprehensive deep dive into the technical and institutional landscape of quantum teleportation research in 2025, with special attention to quantum teleportation mechanisms, protocols, hardware, prominent breakthroughs, key applications, leading research centers and companies, scalability issues, and a survey of both global and Australian contributions. Additionally, we integrate the wealth of resources available, including web links to papers, datasets, preprints, labs, and initiative homepages.

Major 2025 Breakthroughs in Quantum Teleportation

Oxford's Distributed Quantum Supercomputer: Teleporting Quantum Gates

In one of the most widely cited physics breakthroughs of 2025, a team at the University of Oxford achieved scalable quantum teleportation of logical quantum gates across a photonic network, effectively wiring together modular quantum computers as a single, cohesive quantum system. Unlike prior efforts that only teleported single qubit states, the Oxford group demonstrated deterministic, repeatable teleportation of two-qubit gates-crucial building blocks of quantum algorithms-between physically separated modules connected by meters of optical fiber. This was achieved with trapped-ion qubits in each module and photonic links for non-local entanglement and gate operations. Notably, they executed Grover’s quantum search algorithm across the distributed modules with a logic gate fidelity of 86% and successfully demonstrated a

71% success rate in algorithmic tasks, a substantial step toward practical, networked quantum computing[2][4].
This architecture addresses the notorious scalability problem in quantum computing: rather than confining millions of fragile qubits in a monolithic device (which introduces debilitating error rates and engineering complexity), the Oxford team's approach links smaller devices into a scalable network, distributing computational workloads and preserving quantum coherence over longer spatial and temporal scales. Their technique lays vital groundwork for a quantum internet-a global network where secure quantum communication, distributed computation, and remote sensing are not only theorized, but demonstrably feasible[3][5].

Teleportation Over Existing Telecom Cables

Meanwhile, researchers at Northwestern University marked a complementary milestone by successfully achieving quantum teleportation of data across ordinary, commercial internet fiber optic cables-demonstrating quantum signals can coexist with classical internet traffic on the same infrastructure. This test ran quantum teleportation over a live, 30-kilometer network segment, with engineered protocols to separate "quantum" photon signals from the noise of trillions of classical photons. Crucially, it showed that quantum networking need not require entirely new cable deployments; existing infrastructure could be made quantum-compatible, greatly lowering costs and accelerating real-world rollout of quantum internet services[6].

Telecom-Wavelength Quantum Teleportation with Solid-State Quantum Memory

Further East, a quantum photonics team at Nanjing University published a pivotal demonstration of quantum teleportation from a telecom-wavelength photonic qubit to a solid-state erbium-ion quantum memory. Both the entangled source and the memory operated natively at telecommunication C-band (1.5 μm), making the experiment directly compatible with globally deployed fiber networks. This allows direct teleportation and quantum storage of photonic qubits over long distances with high fidelity (exceeding classical limits by over seven standard deviations), overcoming a major hurdle toward scalable, long-lived quantum entanglement distribution for quantum networks. The fidelity reached 81.8% for state teleportation and 73.6% for process teleportation-crucially both above the classical threshold-and the system was shown to be extensible for quantum repeaters and long-distance quantum key distribution[8][10].

Turning Noise into an Ally: Hybrid Entanglement

Breakthroughs in Finland and China demonstrated the transformative impact of so-called "hybrid entanglement," where quantum entanglement is distributed across distinct physical properties (e.g. polarization and frequency of photons) rather than a single mode. This architecture showed, paradoxically, that adding noise can enhance the fidelity of the teleportation process, rather than degrade it. By distributing entanglement between these hybrid degrees of freedom, the teams achieved near-flawless quantum state transfer in noisy environments. This development could enable much more robust quantum communication links in real-world conditions, making quantum networks practical even outside of pristine lab settings[11].

Quantum Teleportation Mechanisms: Entanglement, Superposition, and the Teleportation Protocol

Fundamental Mechanisms

At its core, every quantum teleportation procedure leverages two hallmark features of quantum mechanics: **entanglement** and **superposition**.

- **Superposition**: A qubit can exist in a superposed state-a complex, probabilistic mixture of 0 and 1-until measured, as opposed to classical bits with definite values.
- **Entanglement**: When two (or more) qubits become entangled, the state of one immediately determines the outcome of the other, no matter the distance, creating a non-classical correlation.

Teleportation does not transmit matter or energy in the classical sense. Instead, quantum teleportation allows an **unknown quantum state** to be transferred from a sender (Alice) to a receiver (Bob), relying only on pre-shared entangled pairs and classical communication[13].

Protocols and Quantum Circuit Operations

A standard (single-qubit) quantum teleportation protocol operates as follows:

1. **Entanglement Preparation**: Alice and Bob share a pair of qubits (A and B) prepared in a maximally entangled state (an e-bit).
2. **State Preparation**: Alice possesses an independent, unknown qubit Q (to be teleported).
3. **Bell-State Measurement**: Alice performs a joint, entangling measurement (often a Bell state measurement) on her qubits (A and Q). This projects A and Q onto one of four entangled basis states, collapsing the system’s quantum information accordingly.
4. **Classical Communication**: She sends the result (two classical bits) to Bob.
5. **Conditional Operation**: Bob applies one of four corrective quantum operations on his qubit B, based on Alice’s measurement.
6. **Result**: Bob’s qubit B is now in the original quantum state of Alice’s Q (up to known unitary transformations). The original state at Alice is destroyed, in accord with the no-cloning theorem.

Mathematical Description: The protocol can be written as:

- Initial state:
- Entangled state:
- Alice’s operations: CNOT and Hadamard gates on A and Q, measurement, classical communication (bits a and b)
- Bob’s corrections: if (a, b) = (00), do nothing; (01), apply Z; (10), apply X; (11), apply ZX (Pauli gates).

This protocol is extensible: it can teleport not just isolated qubits, but also entangled states of several qubits, enabling “gate teleportation” and distributed quantum computation[14][15].

Quantum Teleportation Protocols and Advanced Schemes

Numerous protocol variants and experimental schemes have been developed:

Two-Way and Multidirectional Teleportation

Recent protocols enable **two-way** or **bidirectional** teleportation, where two parties can swap quantum states simultaneously. Using six-qubit “cluster states,” researchers have implemented protocols allowing Alice and Bob to teleport arbitrary two-qubit states to one another without needing auxiliary qubits or complex multi-qubit gates. This framework is also extendable to controlled (supervised by a third party), cyclic, and even multiparty teleportation schemes-cornerstones for future quantum internet and conference key sharing applications[15][16].

Analog Quantum Teleportation

Another forward-looking protocol, so-called **analog quantum teleportation**, replaces classical communication with a noisy quantum channel, allowing improved performance depending on channel transmissivity-especially relevant for microwave links in superconducting circuits[17].

Hardware Platforms for Quantum Teleportation

Quantum teleportation has been demonstrated and is being pursued across a plurality of hardware platforms, each with unique strengths and engineering challenges.

Photonic Systems

Photons (particles of light) are ideal for long-distance quantum communication, given their immunity to environmental decoherence and compatibility with fiber optics.

- **Platforms**: Integrated silicon nitride (SiN) photonic chips, cavity-enhanced spontaneous parametric down conversion (SPDC), and telecom-band single-photon sources.
- **Examples**: The Nanjing group’s use of chip-scale resonators and erbium/Yb-doped crystals for quantum memory[9][18].
- **Advantages**: Telecom-band operation-direct compatibility with real fiber-optic telecom networks; proven scalability for quantum key distribution and networked teleportation.
- **Key Experiments**:
  - Quantum teleportation of qubits encoded as time-bin states from photonic sources to rare-earth doped crystals (long storage times, high fidelity)[9].
  - 30 km quantum teleportation over metropolitan fiber with coexistent classical traffic[6].

Trapped Ions

Trapped ion systems offer high-fidelity quantum logic and long coherence times.

- **Oxford Breakthrough**: Utilized small modules of trapped-ion qubits networked by photons through optical fibers. Achieved deterministic teleportation of quantum gates (not just states), validated by executing Grover's algorithm across the distributed modules[2][4].

Superconducting Qubits

Superconducting circuits allow nanosecond gate speeds and easy integration into microwave infrastructure.

- **IBM, Google, Rigetti**: Public quantum devices support circuit-based teleportation experiments, error correction schemes, and even first demonstrations of quantum energy teleportation.
- **Quantum Energy Teleportation (QET)**: Demonstrations have been realized on IBM's quantum hardware. These circuits require precise timing and calibration, and showcase the use of teleportation protocols for remote energy transfer and algorithmic cooling[20][21].

Neutral Atom and Solid-State Systems

- **Neutral Atom Arrays**: Used for logic operations and scalable, parallel qubit arrangements.
- **Solid-State Spin Quantum Memories**: Rare-earth-doped crystals (e.g., Praseodymium, Erbium ions) offer long quantum memory times and compatibility with photonic interfaces[10].

Satellite and Free-Space Channels

Pioneering quantum teleportation over hundreds to thousands of kilometers between satellites and ground stations has been demonstrated, opening the path for intercontinental quantum links and eventually a space-based quantum internet[22].

Quantum Teleportation Protocols and Technology Comparison Table

Hardware Platform	Use Case	Scalability Potential	Demonstrated Distance	Major Labs/Institutions
Integrated Photonics	Long-distance, telecom-band QKD	High	30-1200 km	Nanjing U., USTC, ICFO, Northwestern, Aegiq
Trapped Ions	Distributed QC, gate teleportation	Modular/High	2 m lab, extensible	Oxford, IonQ, Quantinuum, Oxford Ionics
Superconducting Qubits	Teleportation, QET, circuit QC	Medium-High	On-chip to short	IBM, Google, Rigetti, Zurich, Intel

Satellite Quantum Comm	Global QKD, QSC, time transfer	Global/Scalable	500-1400 km	CAS (China), Jinan, European Space Programs
Neutral Atom Arrays	Scalable QC	High, Emerging	On-chip to lab-scale	Pasqal, Atom Computing, Harvard-MIT
Quantum Annealers	Optimization, hybrid quantum net	Niche, high error	On-chip	D-Wave

Each platform presents trade-offs in terms of fidelity, network integration, physical infrastructure, and scalability paths, with ongoing research focusing on converging these technologies for hybrid quantum networks[24].

Distributed Quantum Computing and the Quantum Internet

Modular Architectures and Photonic Interconnects

Distributed quantum computing via teleportation overcomes the scalability “wall” of monolithic architectures, allowing the effective "wiring together" of small quantum modules. The Oxford team’s network of small trapped-ion quantum computers-linked by photons-neatly exemplifies this principle. This enables both computation and memory to be flexibly partitioned and recombined, allowing for infinite, theoretically, system expansion[1][2].

Quantum Repeaters and Quantum Memory

As with classical internet, signal loss and noise present growing obstacles at scale. Quantum repeaters-nodes that receive, store, and forward quantum information via teleportation-are essential for building true long-distance quantum networks. The use of solid-state quantum memories (e.g., doped crystals) at telecom wavelengths facilitates storage and entanglement distribution compatible with deployed infrastructure, crucial for scaling beyond laboratory networks to regional or global quantum internet poles[18][8].

Integration With Existing Infrastructure

Experiments over live fiber-optic cables (with coexistent classical traffic) show that quantum internet deployment need not wait for global top-down re-cabling: existing lines can be harnessed, rapidly accelerating quantum network deployment and reducing costs[25].

Secure Communications and Quantum Teleportation

Quantum teleportation provides the foundation for provably secure communications systems:

- **Quantum Key Distribution (QKD)**: Entanglement-based QKD protocols use teleportation to deliver encryption keys that cannot be intercepted or cloned, ensuring data confidentiality even against quantum computer attacks[22][7].

- **Quantum Cryptography**: Teleportation secrets cannot be eavesdropped without disturbing the system (no-cloning theorem). Any interception is instantly detectable by the correlated parties.
- **Multi-Party Communication**: Bidirectional and controlled teleportation schemes allow sharing of encryption keys and data across multiple users and nodes-enabling conference key sharing and multi-user QKD.
- **Quantum Sensing and Timing**: Quantum networks employing teleportation enable quantum clock synchronization and distributed sensing for defense, navigation, and financial applications. Australian researchers, especially at the University of Adelaide, are leading efforts on deployable quantum clocks and quantum-secured navigation[28].

Quantum Teleportation and Error Correction

While foundational, quantum teleportation-like all quantum protocols-remains vulnerable to noise and errors (decoherence, gate errors, and photon loss). Addressing fault tolerance and error correction is paramount.

- **Teleportation-Based Error Correction**: Certain quantum error correction codes employ teleportation as a means of moving quantum information between different parts of a computer, or transferring error syndromes without destroying data qubits. Examples include surface codes and measurement-based quantum computing (MBQC), with demonstrated protocols on IBM systems[30][31].
- **Scheduling and Syndrome Policies**: Advanced scheduling strategies (e.g., Freshest Qubit First) optimize which qubits to teleport first based on error likelihood, significantly improving fidelity in noisy or high-load quantum networks.
- **Resource Overhead**: Error correction and teleportation are resource-intensive, often requiring additional qubits and circuit depth, challenging quantum hardware to scale fast enough while maintaining low error rates[31].

Emerging Applications: Beyond Communication and Computing

Quantum Sensing and Distributed Measurement

Quantum teleportation underpins distributed sensing applications, enabling ultra-precise, remote-synchronized measurements (crucial for atomic clocks, geodesy, and gravitational wave detection). Quantum networks built with teleportation protocols could one day synchronize global time standards and power ultra-sensitive scientific sensor arrays[32][7].

Quantum Energy Teleportation (QET)

Emerging research demonstrates quantum energy teleportation-using local operations and classical communication on entangled systems to transfer energy across spatial gaps, without conventional carriers. Experimental validations on IBM's superconducting quantum hardware and NMR systems show applications in algorithmic cooling (reducing entropy of subsystems), generating negative energy densities (relevant for semi-classical gravity), and possibly new approaches to energy transfer and quantum engines[33][20].

Global Research Institutions Leading Quantum Teleportation

The progress seen in 2025 is the product of a dynamic, multi-institutional ecosystem. The table below highlights key centers, their focus areas, and notable achievements:

Institution/Lab	Country	Primary Focus areas	Milestone Achievements	Resource Link(s)
University of Oxford Dept. of Physics & QCS Hub	UK	Photonic networks, trapped-ion QC	Distributed quantum gate teleportation	Oxford QCS Hub
MaLab, Nanjing University	China	Telecom quantum memory, photonic chips	First telecom-band quantum memory teleport	MaLab, Phys. Rev. Lett. 135, 010804
ICFO, Barcelona	Spain	Photonic memories, multiplexed teleport	Long-distance, multiplexed teleportation	Nature Comm.
Aegiq (ex-Shef field)	UK	Photonic quantum networks /computing	Full-stack photonic QC + QKD	Aegiq
IBM Quantum	USA	Superconducting qubits, cloud QC	Teleportation circuits, QET, error correction	IBM Quantum Learning
Google Quantum AI	USA	Superconducting QC, wormhole simulation	Teleportation, distributed logic	Quantum AI
Rigetti, Intel, D-Wave, Quantinu um, IonQ	USA/EU/UK	Multi-platform quantum devices	Hardware implementations, distributed QC	Rigetti, IonQ

Chinese Academy of Sciences (CAS, USTC)	China	Satellite quantum networks, QKD	Space-ground teleportation, “Micius”	Satellite-Relayed Quantum Net
University of Adelaide IPAS, QuantX	Australia	Quantum sensing/clocks, industry links	Deployable quantum clocks, time transfer	IPAS, QuantX Labs

Australian institutions actively collaborate with global leaders and act as a southern hemisphere hub, especially in photonics, quantum sensing, and precision measurement[32][28].

Australian Contributions to Quantum Teleportation Research

Australia is a prominent contributor to global quantum research, with active centers, multi-million-dollar infrastructure projects, and public-private partnerships.

University of Adelaide (IPAS, CQMQT)

- **Infrastructure**: World-class labs for quantum materials, device fabrication, and ultrastable quantum sensing[32].
- **Quantum Australia**: Collaborative national program for quantum technology leadership.
- **Quantum Sensing and Secure Timing**: Quantum clocks for defense, telecommunications, and navigation-including upcoming satellite launches to test secure, quantum-based time standards[27].

Australian Satellite Quantum Experiments

Australian researchers are engaged in CubeSat-based quantum communication missions, like the CQuCoM experiment, testing entangled photon sources and quantum key distribution from low Earth orbit, partnering with Asian and European teams. These projects aim to establish sovereign, space-based quantum communication capacity[22].

Industry Partnerships

Firms like QuantX Labs, spun out from the University of Adelaide, have received major defense and technology grants to develop field-ready quantum clocks and quantum-secured time transfer systems-a key leap toward quantum-enhanced navigation and precise coordination for military, space, and financial system applications[27].

Technical Challenges and Scalability Solutions

Despite great strides, significant technical hurdles remain:

- **Decoherence and Fidelity Loss**: Preserving entanglement over long distances and times remains a core challenge, as noise quickly destroys quantum correlations.

- **Integration and Hybrid Networks**: Bridging quantum and classical networks, and combining photonic, ion, atom, and superconducting platforms, requires new protocols and hardware engineering.
- **Error Correction Overhead**: Advanced error correcting codes dramatically increase hardware requirements, but are indispensable for scaling up beyond hundreds of qubits.
- **Quantum Repeaters**: Efficient, high-fidelity quantum repeater designs are needed for continental and global scale networking.
- **Resource Scheduling**: As demonstrated by the FQF (Freshest Qubit First) policy, intelligent resource scheduling and error syndrome tracking markedly improve network fidelity, but optimizing these policies remains a subject of research[31].
- **Hardware Manufacturability**: The ability to produce, maintain, and operate quantum processors at scale relies on logistical advances in cryogenics, photonic chip foundry manufacturing, and control electronics[24].

Major consortia and public repositories document, track, and compare roadmap milestones and hardware developments:

- Quantum Hardware Companies and Roadmaps 2025 Edition
- IBM Quantum Roadmap

Comprehensive Resource Compendium

Below is a selection of resources, papers, and lab pages for further exploration, with a focus on recent breakthroughs, protocols, hardware, error correction, and key labs:

Resource/Paper	Link	Focus Area
Oxford distributed quantum algorithm (Nature, 2025)	Nature (2025)	Gate teleportation, modular QC
Nanjing University telecom-band memory teleportation	Phys. Rev. Lett. 135, 010804	Telecom-band QC memory
Review of Quantum Energy Teleportation	arXiv:2505.04689	QET, applications
Demonstration of Quantum Energy Teleportation on IBM QC	Phys. Rev. Applied 20, 024051	QET on superconducting hardware
Enhanced Two-Way Teleportation Protocol	arXiv:2412.21179	Two-way, bidirectional teleportation
Bidirectional Controlled Teleportation Cluster-State	Springer	Protocol, six-qubit cluster state
ICFO multiplexed teleportation	Nature Communications	Photonic memory, long-distance QC
MaLab - Nanjing University	MaLab	Photonic quantum networks

IBM Quantum Teleportation Course	[IBM] (https://quantum-computing.ibm.com/learn/quantum-teleportation)	Teleportation, basic circuits
PostQuantum hardware vendor comparison	PostQuantum	Roadmap, platform assessment
Australian Centre for Quantum Growth	CQG	National quantum research
Institute for Photonics and Advanced Sensing (IPAS)	IPAS	Quantum sensing, Adelaide
QuantX Labs	QuantX Labs	Precision quantum clocks

Conclusion

The state of quantum teleportation research in 2025 is dynamic, robust, and converging rapidly toward global impact. From laboratory curiosities to practical demonstrations spanning photonic chips, trapped ions, and real-world fiber infrastructure, quantum teleportation is reshaping the limits of secure communications, distributed quantum computing, and time-and-position sensing. The successes of the Oxford distributed quantum supercomputer and Nanjing's telecom-band memory teleportation serve as harbingers for a coming era of functional quantum networks-where computation, communication, and remote measurement are conducted in ways once unimaginable.

The technical challenges are formidable, notably error correction, environmental noise, and hardware scaling. Yet, continued progress in quantum repeater technologies, multiplexed photon sources, integrated photonics, and error correction schemes point toward practical quantum internet within the decade. Australia, through its advanced quantum sensors, collaborative infrastructure, and innovative startups, remains a major contributor to this unfolding narrative, ensuring that the benefits-and competitive edge-of quantum teleportation are woven deeply into its future technology landscape.

As the field surges forward, those interested in contributing, observing, or investing in quantum technologies are encouraged to explore the resources and institutions listed throughout this report, and to track the ever-accelerating rollout of use-case demonstrations, public quantum computing resources, and global quantum networking initiatives. The quantum internet, once science fiction, is rapidly moving toward science fact.

References (33)

1. *Oxford team teleports first quantum gate in landmark paper*. https://www.theregister.com/2025/02/12/oxford_quantum_gate_teleportation/

2. *Oxford Researchers Demonstrate Distributed Quantum Computing via a …*. https://quantumcomputingreport.com/oxford-researchers-demonstrate-distributed-quantum-computing-via-a-photonic-network/
3. *First distributed quantum algorithm brings quantum supercomputers …*. https://www.ox.ac.uk/news/2025-02-06-first-distributed-quantum-algorithm-brings-quantum-supercomputers-closer
4. *Oxford scientists achieve teleportation with quantum supercomputer*. https://www.independent.co.uk/news/science/oxford-teleportation-quantum-computing-supercomputer-b2698105.html
5. *The Star Trek Era Begins: Quantum Teleportation Becomes Reality in 2025*. https://discoverwildscience.com/the-star-trek-era-begins-quantum-teleportation-becomes-reality-in-2025-1-335722/
6. *Quantum teleportation makes big leap in "flawless" new experiment …*. https://www.earth.com/news/quantum-teleportation-takes-big-leap-forward-new-experiment/
7. *Quantum internet moves closer as researchers teleport light-based …*. https://phys.org/news/2025-07-quantum-internet-closer-teleport-based.html
8. *Quantum Teleportation from Telecom Photons to Erbium-ion Ensembles*. https://arxiv.org/abs/2505.05233
9. *Quantum teleportation* . https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information/entanglement-in-action/quantum-teleportation
10. *[2412.21179] Enhanced Two-Way Teleportation of Entangled States with …*. https://arxiv.org/abs/2412.21179
11. *Bidirectional Controlled Teleportation via Six-Qubit Cluster State*. https://link.springer.com/article/10.1007/s10773-013-1694-0
12. *Cyclic quantum teleportation of two-qubit entangled states by using six …*. https://https-www-nature-com-443.webvpn.ynu.edu.cn/articles/s41598-024-63395-z.pdf
13. *[2502.10253] Analog Quantum Teleportation - arXiv.org*. https://arxiv.org/abs/2502.10253
14. *Quantum Teleportation from Telecom Photons to Erbium-ion Ensembles*. https://arxiv.org/html/2505.05233v1
15. *Long distance multiplexed quantum teleportation from a telecom photon …*. https://www.nature.com/articles/s41467-023-37518-5.pdf
16. *Benchmarking a Quantum Teleportation Protocol in Superconducting …*. https://link.aps.org/doi/10.1103/PhysRevLett.108.040502
17. *Quantum Computing Roadmaps: A Look at The Maps And Predictions of Major …*. https://thequantuminsider.com/2025/05/16/quantum-computing-roadmaps-a-look-at-the-maps-and-predictions-of-major-quantum-players/
18. *[2301.02666] Demonstration of Quantum Energy Teleportation on …*. https://arxiv.org/abs/2301.02666
19. *Application and Development of QKD-Based Quantum Secure … - MDPI*. https://www.mdpi.com/1099-4300/25/4/627

20. *Teleportation Achieved Between Quantum Computers in a World First*. https://www.sciencealert.com/teleportation-achieved-between-quantum-computers-in-a-world-first
21. *Quantum Teleportation 101* . https://www.bluequbit.io/quantum-teleportation
22. *Centre for Quantum Materials & Quantum Technologies* . https://www.adelaide.edu.au/ipas/quantum-materials
23. *NJU - Malab*. https://qoqi.nju.edu.cn/
24. *IPAS Spinoff Goes to Space - The University of Adelaide*. https://www.adelaide.edu.au/ipas/news/list/2025/08/12/ipas-spinoff-goes-to-space
25. *[1902.01692] Demonstration of teleportation-based error correction in …*. https://arxiv.org/abs/1902.01692
26. *Advancements in Quantum Teleportation and Error Correction*. https://scisimple.com/en/articles/2025-06-26-advancements-in-quantum-teleportation-and-error-correction--a9p2ow6
27. *QuantX and University of Adelaide to advance and commercialise quantum …*. https://www.adelaide.edu.au/ipas/news/list/2025/07/14/quantx-and-university-of-adelaide-to-advance-and-commercialise-quantum-clock
28. *[2505.04689] A review of applications of Quantum Energy Teleportation …*. https://arxiv.org/abs/2505.04689